

Data Science and Analytics

Chapter 3 : Data Exploration

1. Introduction

2. Descriptive Statistics

3. Correlation Analysis

4. Categorical Analysis

5. Using Seaborn Library

6. Workshop A: Analyze and Visualize a Marketing Data

7. Conclusions

Data Exploration

- Data Exploration is the initial phase of analysis where raw datasets are examined to uncover structure, detect anomalies, and reveal meaningful patterns.
- It combines statistical methods and visual techniques to ensure data quality and guide model selection. Data exploration transforms raw numbers into insights that are both reliable and actionable.

Chapter's Objectives

- To enable students to summarize data using fundamental descriptive statistics and interpret distributions through histograms and boxplots.
- To help students analyze and visualize relationships between variables using correlation measures, scatter plots, and heatmaps.
- To teach students how to explore categorical data and compare groups effectively using bar charts, stacked bars, and cross-tabulation.

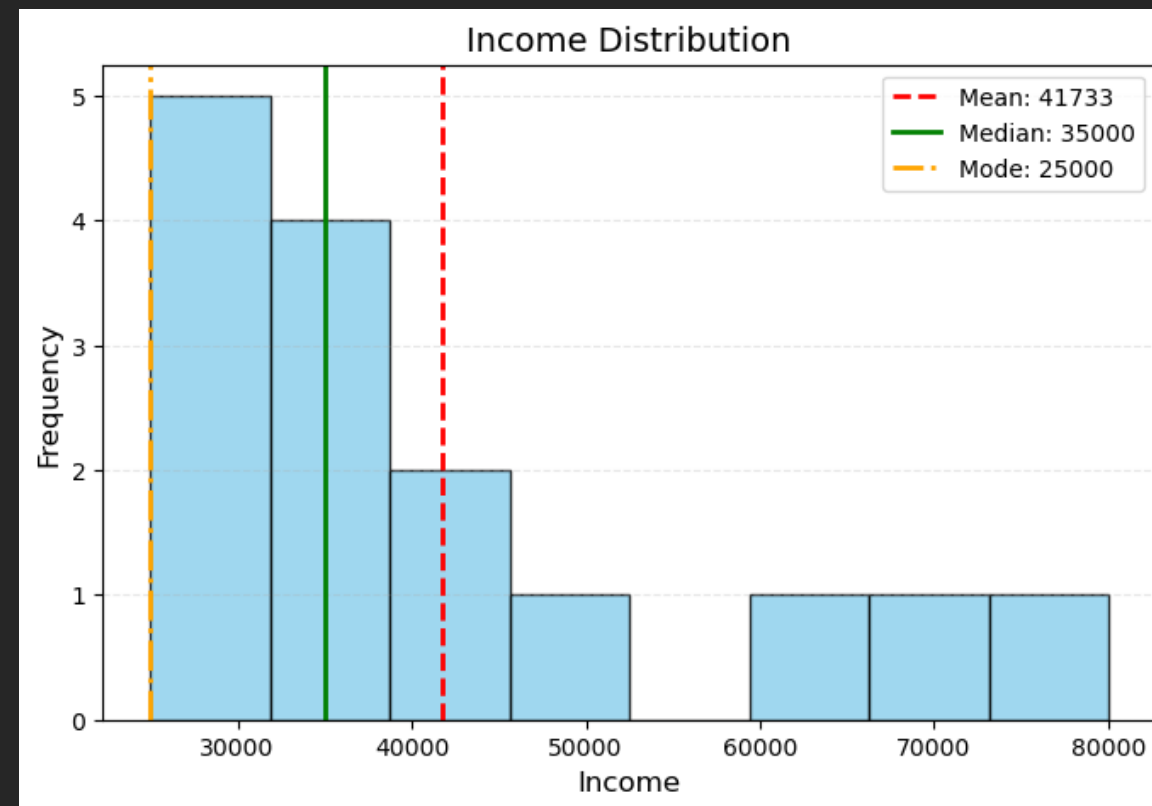
1. Measures of Central Tendency

Measures	Formular	Python
Mean: The arithmetic average of all data points. Sensitive to extreme values (outliers).	$\text{Mean} = \frac{\sum x_i}{n}$	<pre>df["Column"].mean()</pre>
Median: The middle value when data is ordered. More robust against outliers.	$\text{Median} = \begin{cases} x_{\frac{n+1}{2}}, & \text{if } n \text{ is odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}, & \text{if } n \text{ is even} \end{cases}$	<pre>df["Column"].median()</pre>
Mode: The most frequently occurring value. Useful for categorical or skewed data.	Mode=Value with highest frequency	<pre>df["Column"].mode()</pre>

1. Measures of Central Tendency (Cont.)

- Introducing Histogram

Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



** If values are all unique, Mode will return all values (Above display first values)

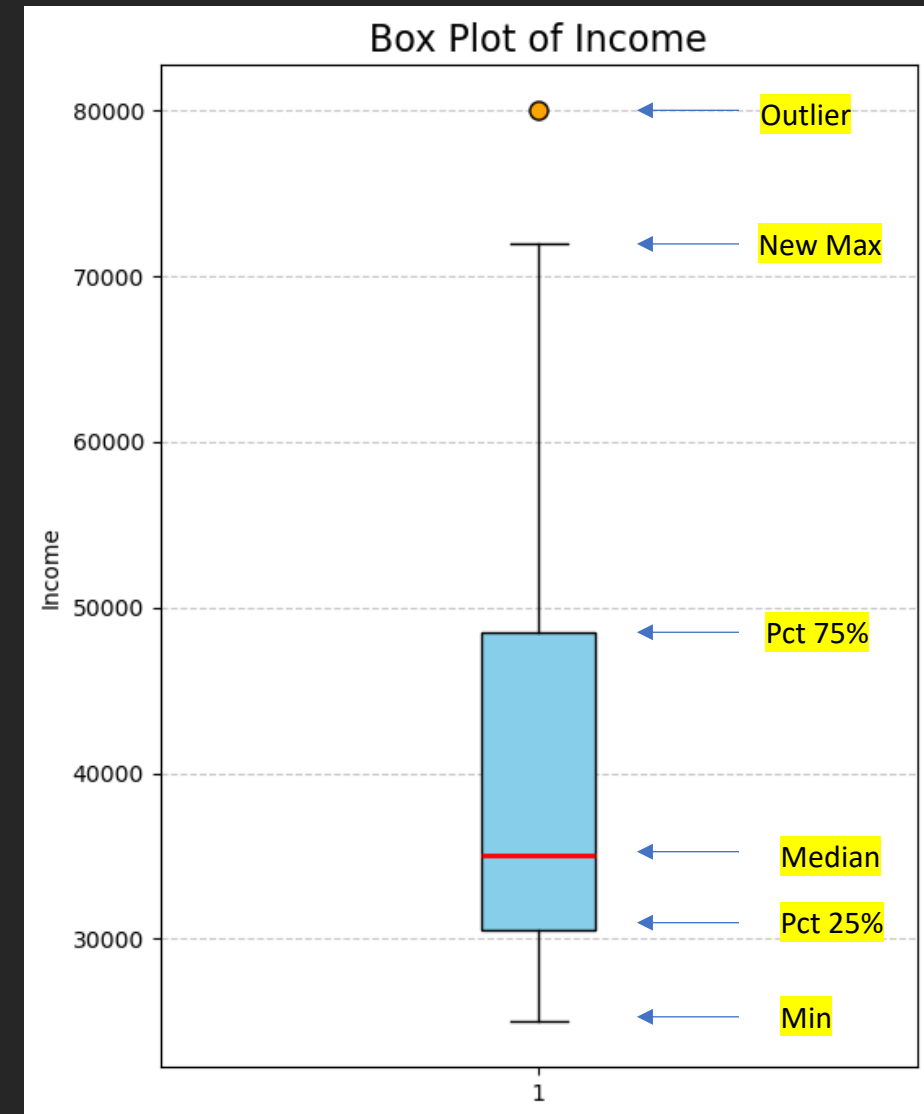
2. Measures of Dispersion

Measures	Formular	Python
Range: The difference between the maximum and minimum values. Shows the total spread of data.	$\text{Range} = \text{Max} - \text{Min}$	<pre>df["Column"].max() - df["Column"].min()</pre>
Variance: The average of squared deviations from the mean. Indicates how far values are spread out.	$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$	<pre>df["Column"].var()</pre>
Standard Deviation: The square root of variance. Expresses typical deviation from the mean in <u>the same units</u> as the data.	$\sigma = \sqrt{\sigma^2}$	<pre>df["Column"].std()</pre>
Inter Quartile Range (IQR): The difference between the Q3 percentile and Q1 percentile. - Shows the spread of the middle 50% of data - Robust against outliers.	$\text{IQR} = Q_3 - Q_1$	<pre>df["Column"].quantile(0.75) - df["Column"].quantile(0.25)</pre>

2. Measures of Dispersion (Cont.)

- Introducing Box Plot

Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



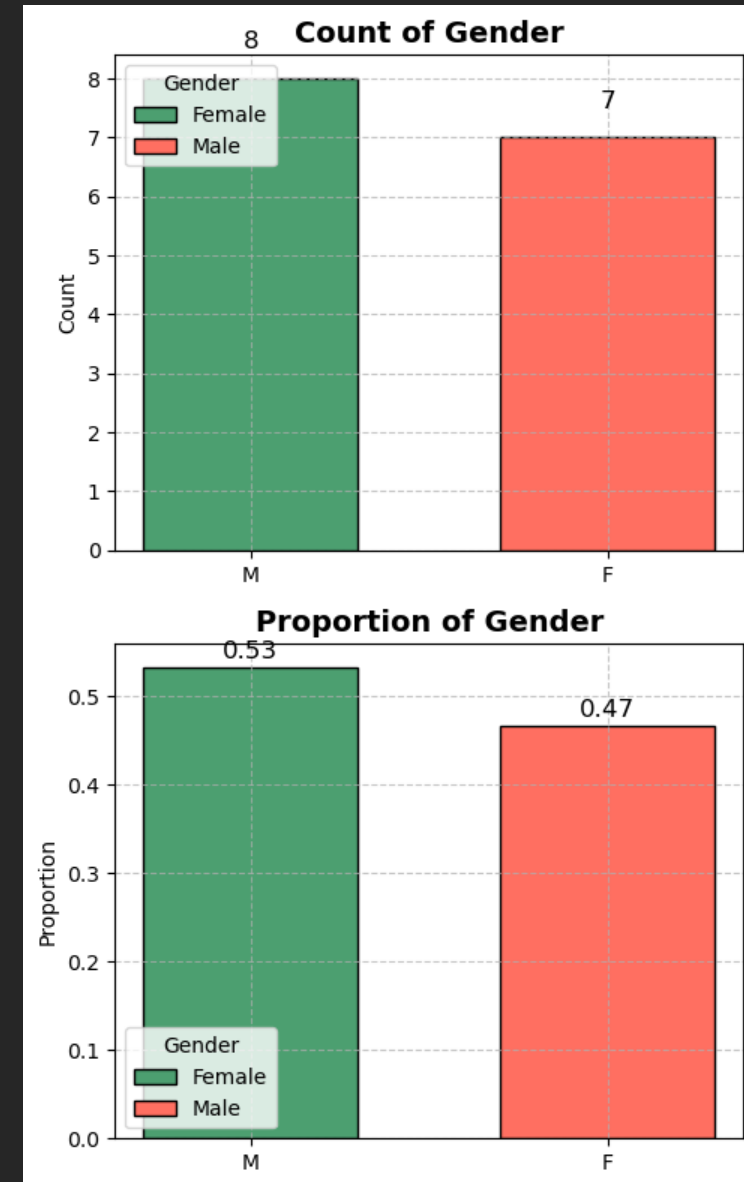
3. Frequency Distribution

Measures	Formular	Python
Count: The number of times a value appears.		<pre>df["Column"].count() df["Column"].value_count()</pre>
Proportion: The fraction of the total represented by each category.	Proportion = $\frac{\text{Count}}{\text{Total}}$	<pre>df["Column"].value_count() / df["Column"].count()</pre>

3. Frequency Distribution (Cont.)

- Introducing Bar Chart

Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



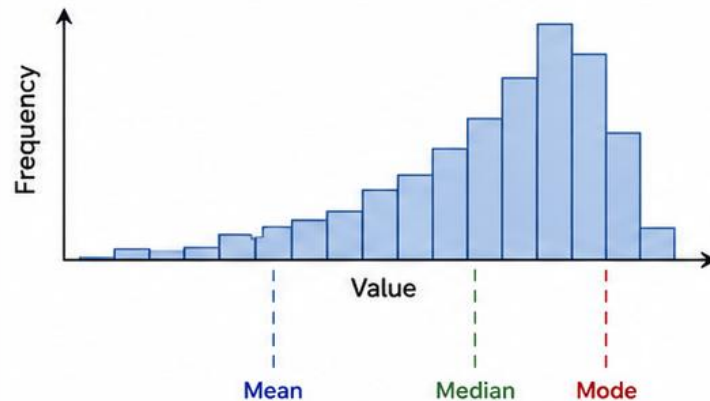
4. Shape of Distribution

Measures	Formular	Python
Skewness: A measure of symmetry • Positive skewness → longer right tail (mean > median). • Negative skewness → longer left tail (mean < median). • Zero skewness → approximately symmetric distribution.	$g_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})^3 / n}{s^3}$	<pre>df["Column"].skew()</pre>
Kurtosis: A measure of tail heaviness relative to a normal distribution. • Mesokurtic (≈ 0) → normal-like tails. • Leptokurtic (> 0) → heavy tails, more extreme values (e.g., financial returns). • Platykurtic (< 0) → light tails, fewer extremes (e.g., uniform distribution).	$g_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^4 / n}{s^4} - 3$	<pre>df["Column"].kurt()</pre>

4. Shape of Distribution (Cont.)

1) Skewed Left (Negative Skew)

Long tail on the **left side**
Most data concentrated on the **right**

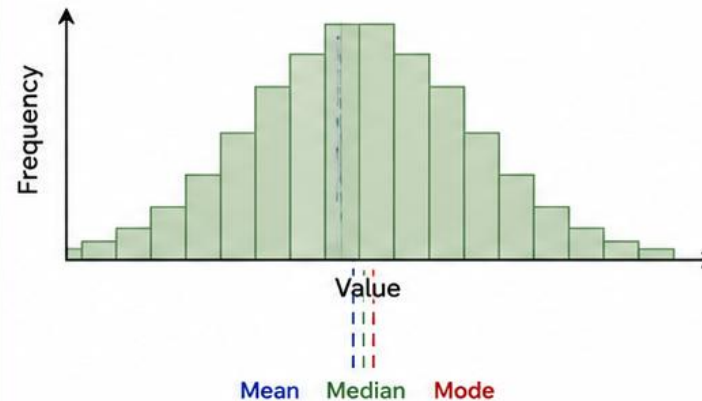


$$\text{Mean} < \text{Median} < \text{Mode}$$

Low values -----> High values

2) Symmetric (No Skew)

Balanced distribution
Data is **evenly distributed** around the center

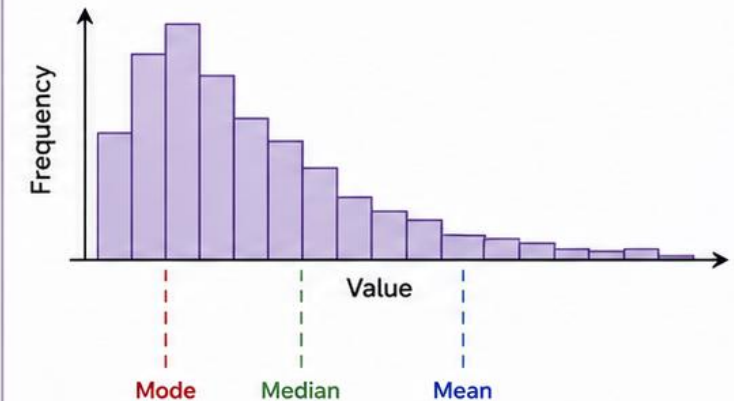


$$\text{Mean} = \text{Median} = \text{Mode}$$

Low values -----> High values

3) Skewed Right (Positive Skew)

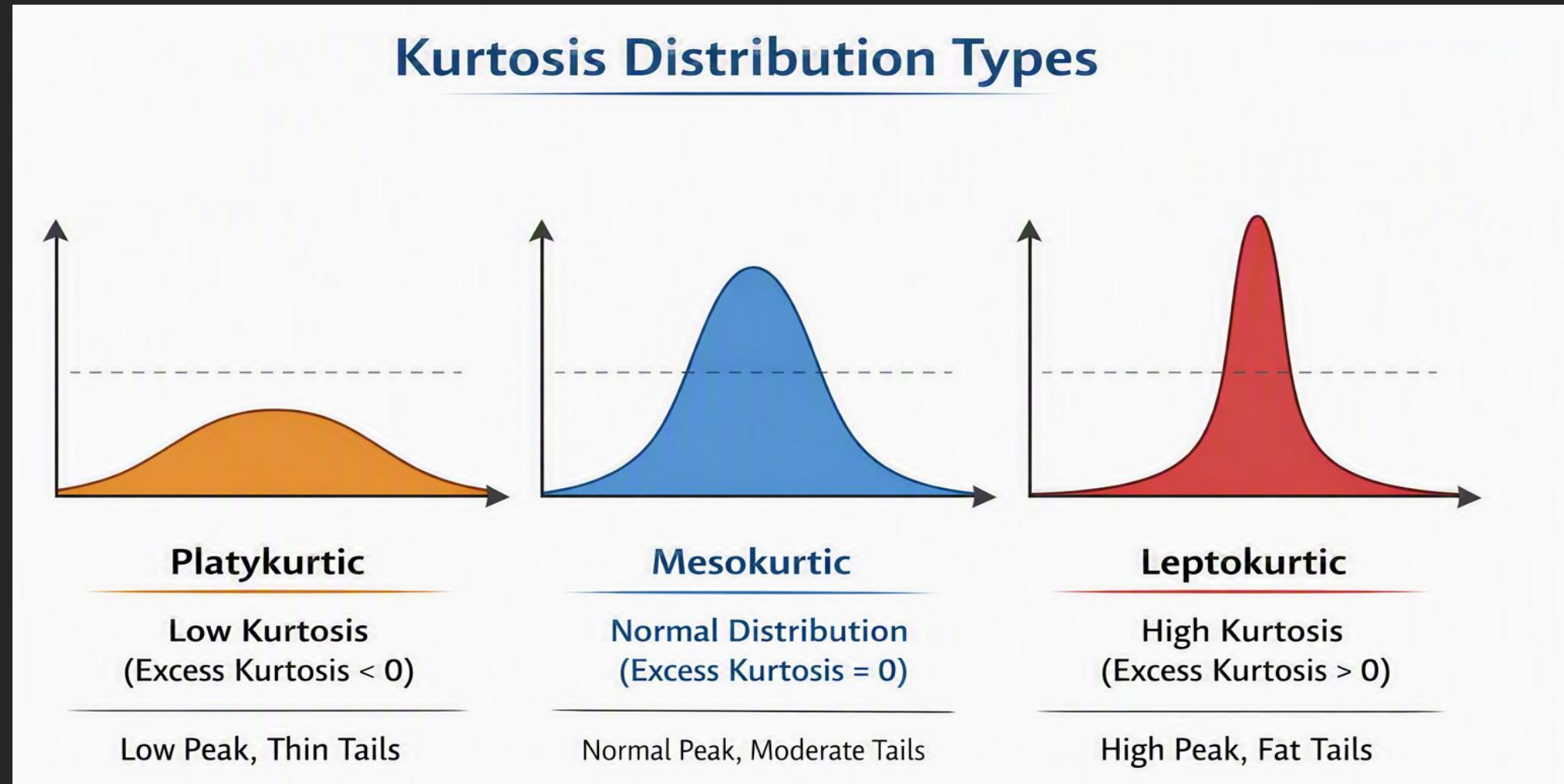
Long tail on the **right side**
Most data concentrated on the **left**



$$\text{Mode} < \text{Median} < \text{Mean}$$

Low values -----> High values

4. Shape of Distribution (Cont.)



Correlation Analysis

1. Correlation

- measures the strength and direction of a linear relationship between two variables. It tells us whether changes in one variable are associated with changes in another.

Type	Description	Example
Positive correlation	Both variables increase together	Height vs. Weight
Negative correlation	One increases while the other decreases	Price vs. Demand
No correlation	No consistent relationship	Shoe size vs. IQ



1. Correlation (Cont.)

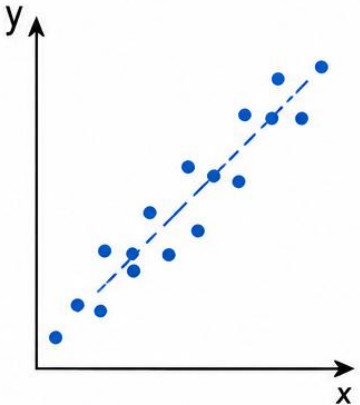
Measures	Formular	Python
Pearson Correlation Coefficient (r): measures linear relationships between continuous variables. Range: $-1 \leq r \leq 1$ • $r \approx 1$ Strong positive correlation • $r \approx -1$ Strong negative correlation • $r \approx 0$ No linear correlation	$r = \frac{\text{Cov}(X, Y)}{S_X \cdot S_Y}$ where: Cov(X, Y) = covariance, S_X, S_Y = standard deviations.	<pre>pearson_r = df["Col1"].corr(df["Col2"], method="pearson")</pre>
Spearman Rank Correlation: measures monotonic relationships when data are ordinal or not normally distributed. Range: $-1 \leq r \leq 1$ • $r \approx 1$ Strong positive correlation • $r \approx -1$ Strong negative correlation • $r \approx 0$ No linear correlation	$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$ where: d_i = difference between ranks of each pair, n = number of observations.	<pre>spearman_r = df["Col1"].corr(df["Col2"], method="spearman")</pre>

1. Correlation (Cont.)

Pearson vs Spearman Correlation

Two common ways to measure association

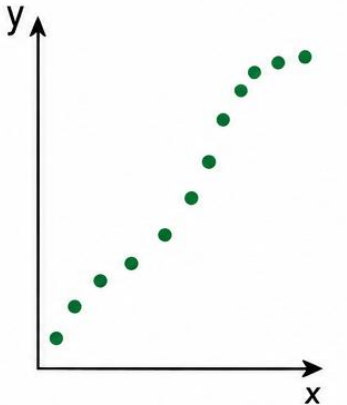
Pearson Correlation



- Measures **linear** relationship
- Uses **raw values**
- Best for **continuous** numeric data
- More sensitive to **outliers**
- Typical symbol: **r**

When to use?
When data are continuous, normally distributed, and the relationship is linear.

Spearman Correlation



- Measures **monotonic** relationship
- Uses **ranks**
- Useful for **ordinal** or **non-normal** data
- Less sensitive to **outliers**
- Typical symbol: **ρ or r_s**

When to use?
When data are ordinal, not normally distributed, or when the relationship is monotonic.

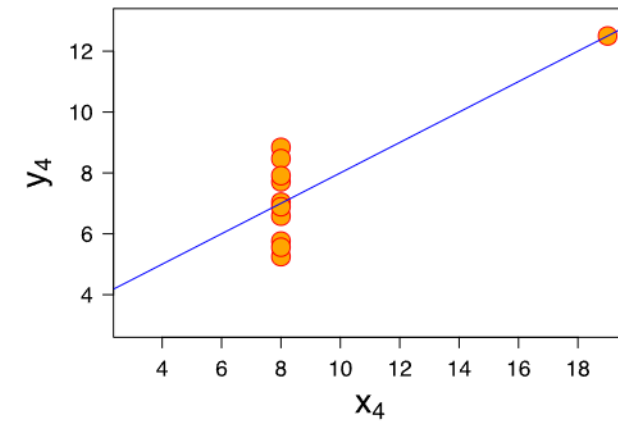
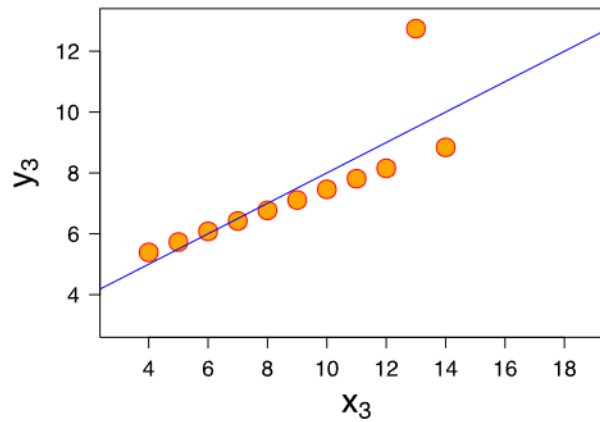
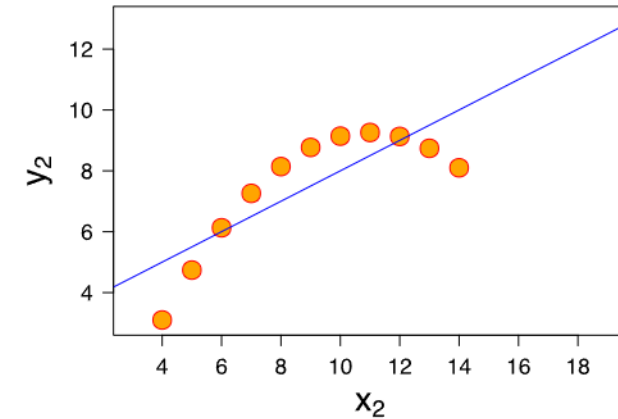
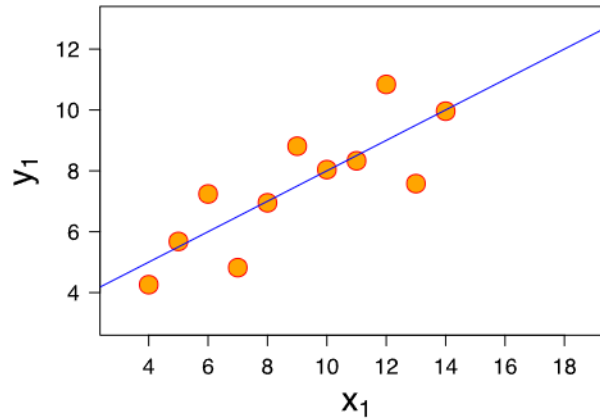
Value (r or ρ)	-1	-0.5	0	0.5	1
Strength	Perfect negative	Moderate negative	No correlation	Moderate positive	Perfect positive

Key Takeaways

- Pearson measures linear relationship using raw values.
- Spearman measures monotonic relationship using ranks.
- Both range from -1 to +1. -1: strong negative 0: no correlation +1: strong positive

1. Correlation (Cont.)

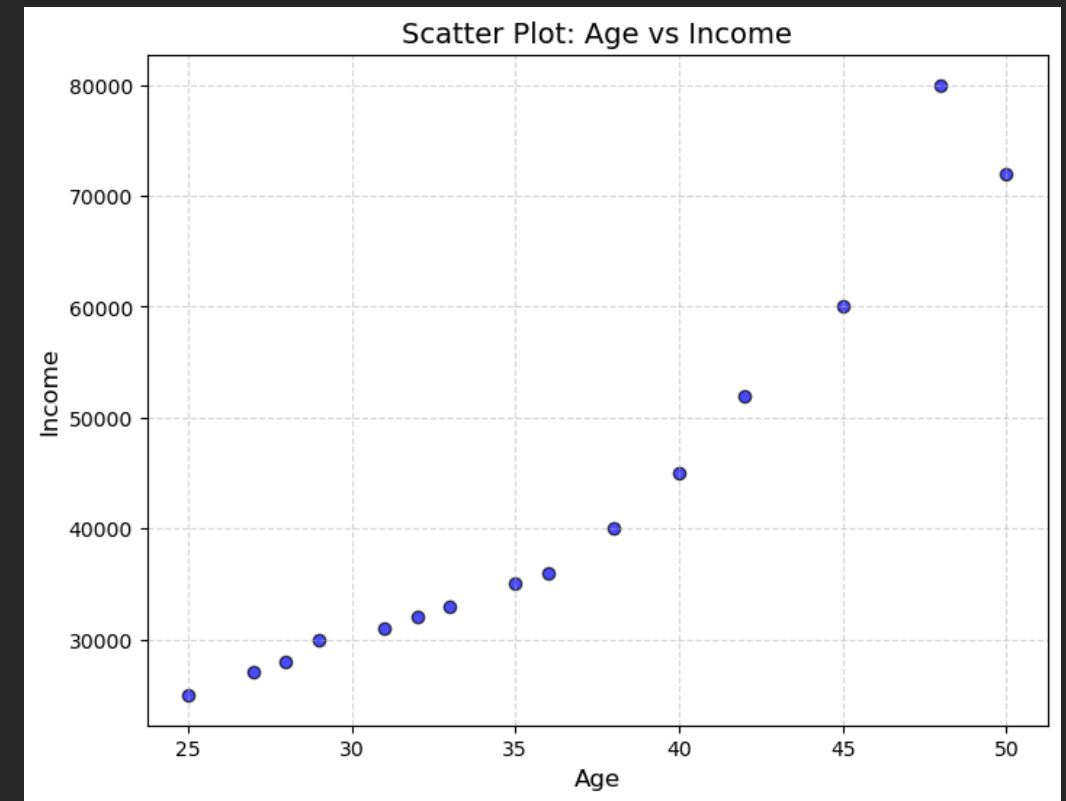
four sets of data with the same correlation of 0.816



1. Correlation (Cont.)

- Introducing Scatter Plot

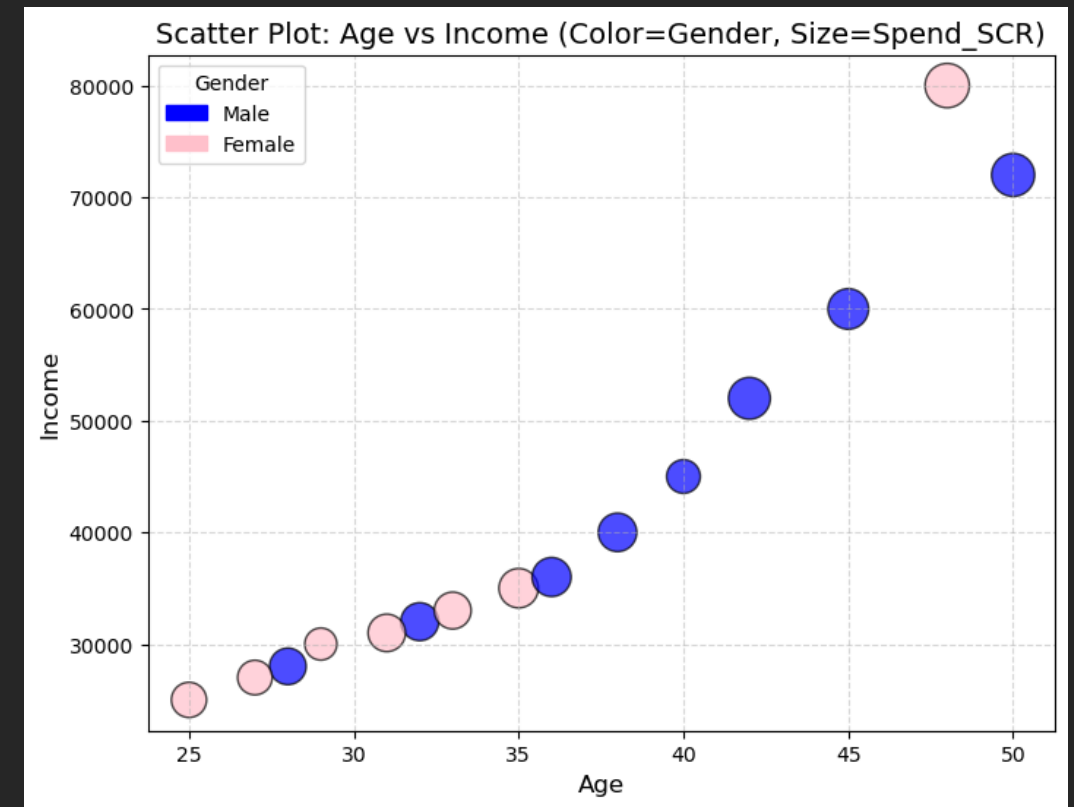
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Correlation (Cont.)

- Introducing Scatter Plot

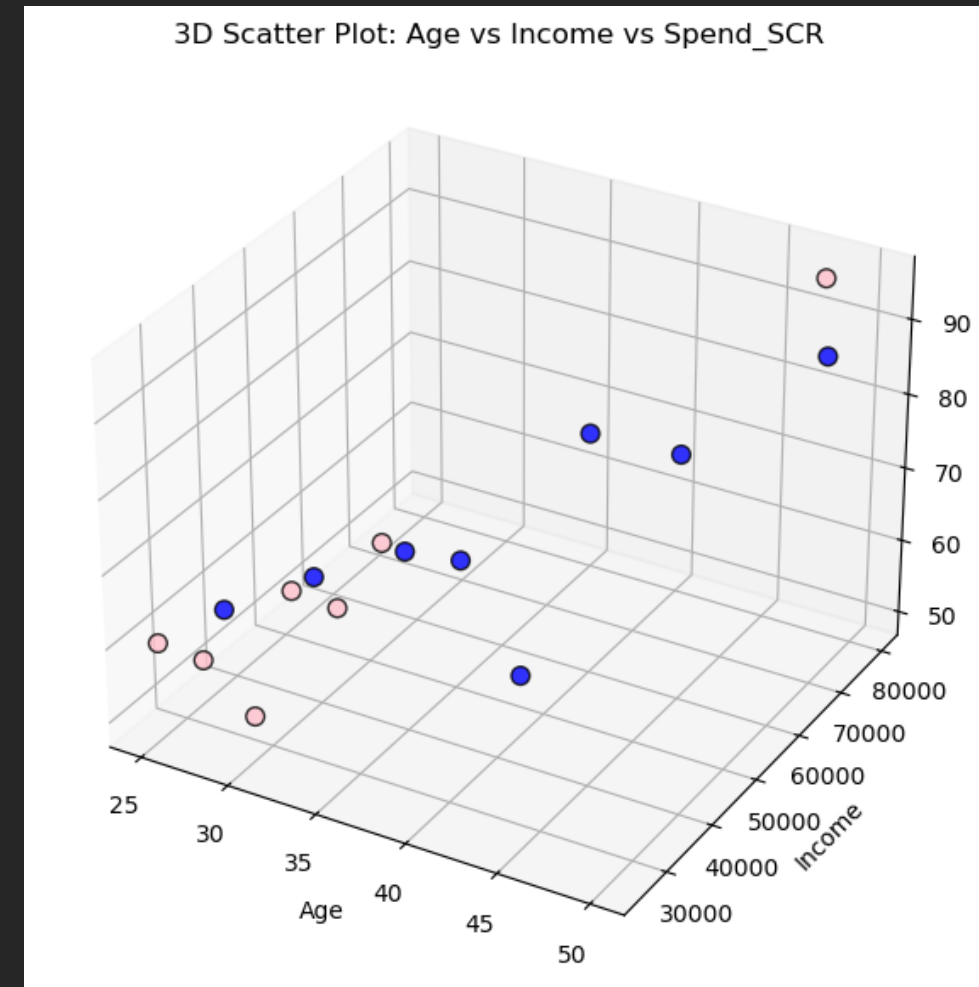
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Correlation (Cont.)

- Introducing Scatter Plot

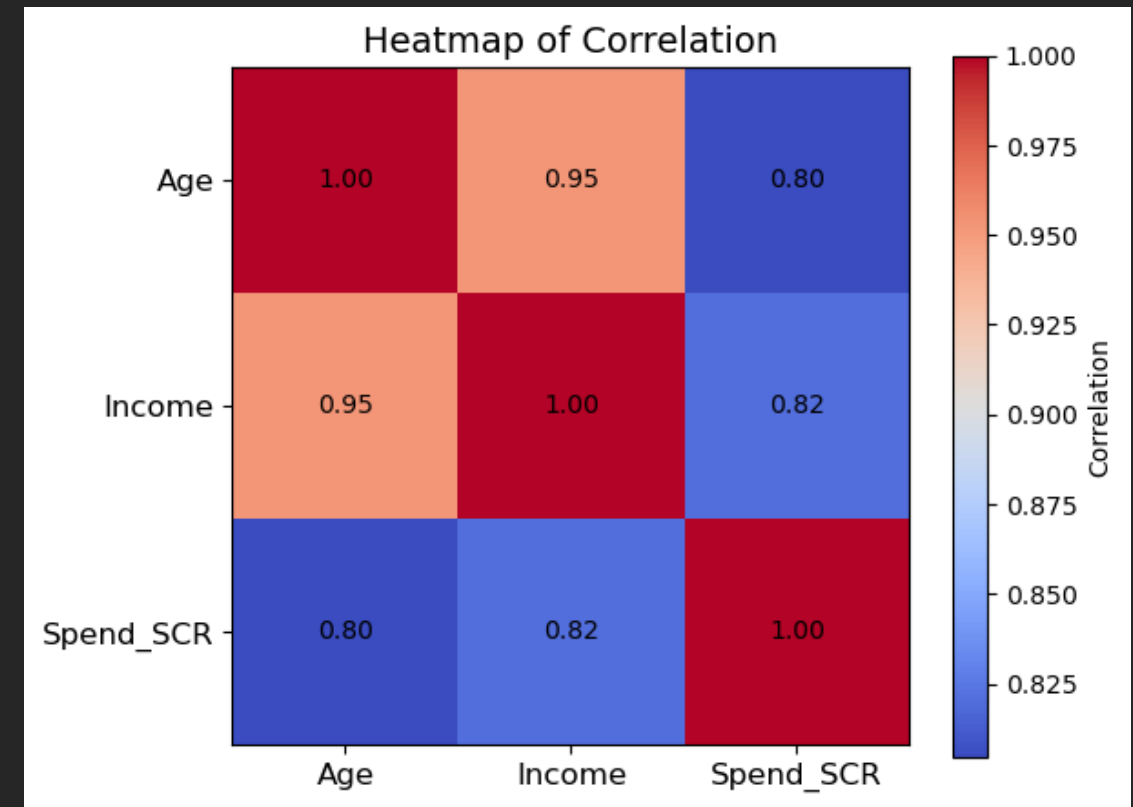
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Correlation (Cont.)

- Introducing Correlation Matrix + Heatmap

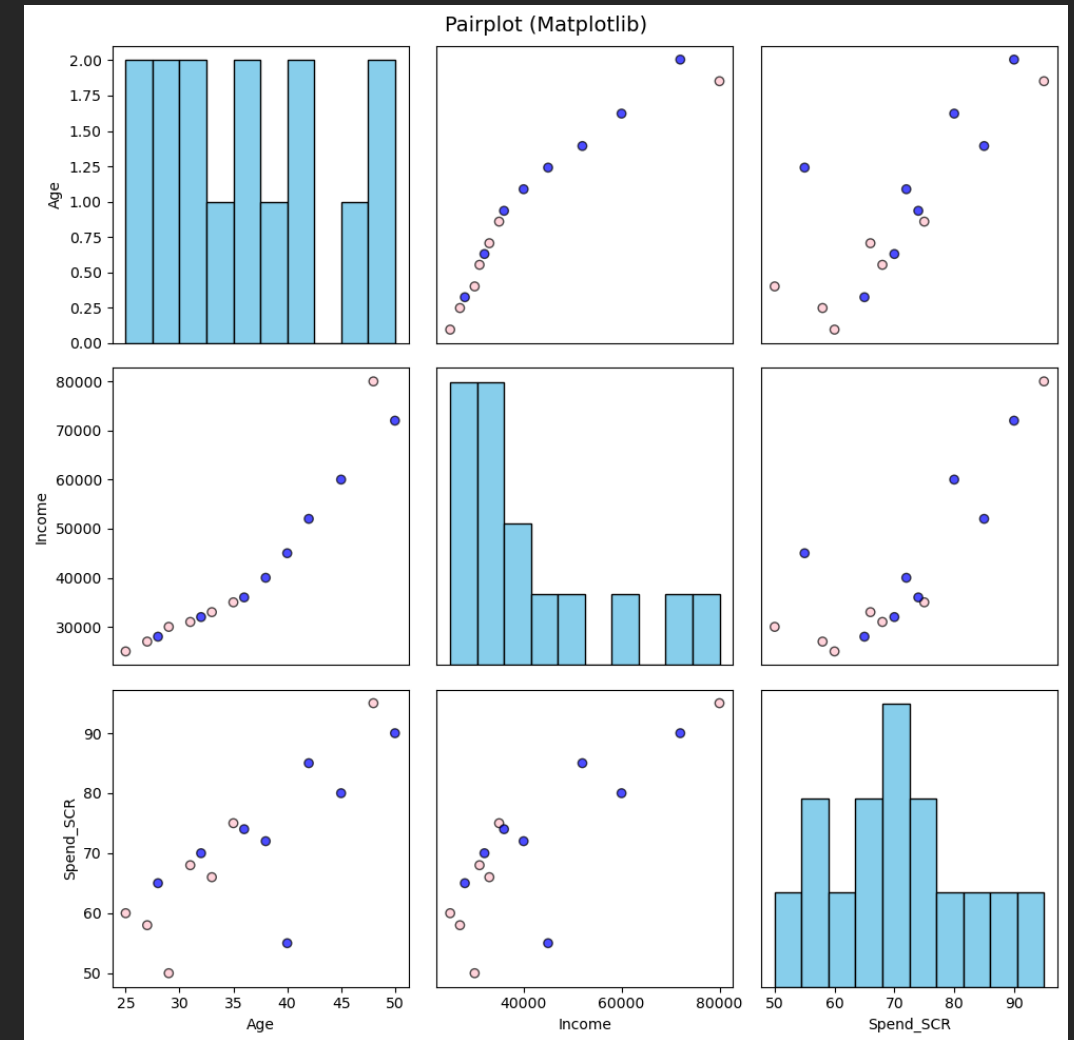
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Correlation (Cont.)

- Introducing Pair-Plot

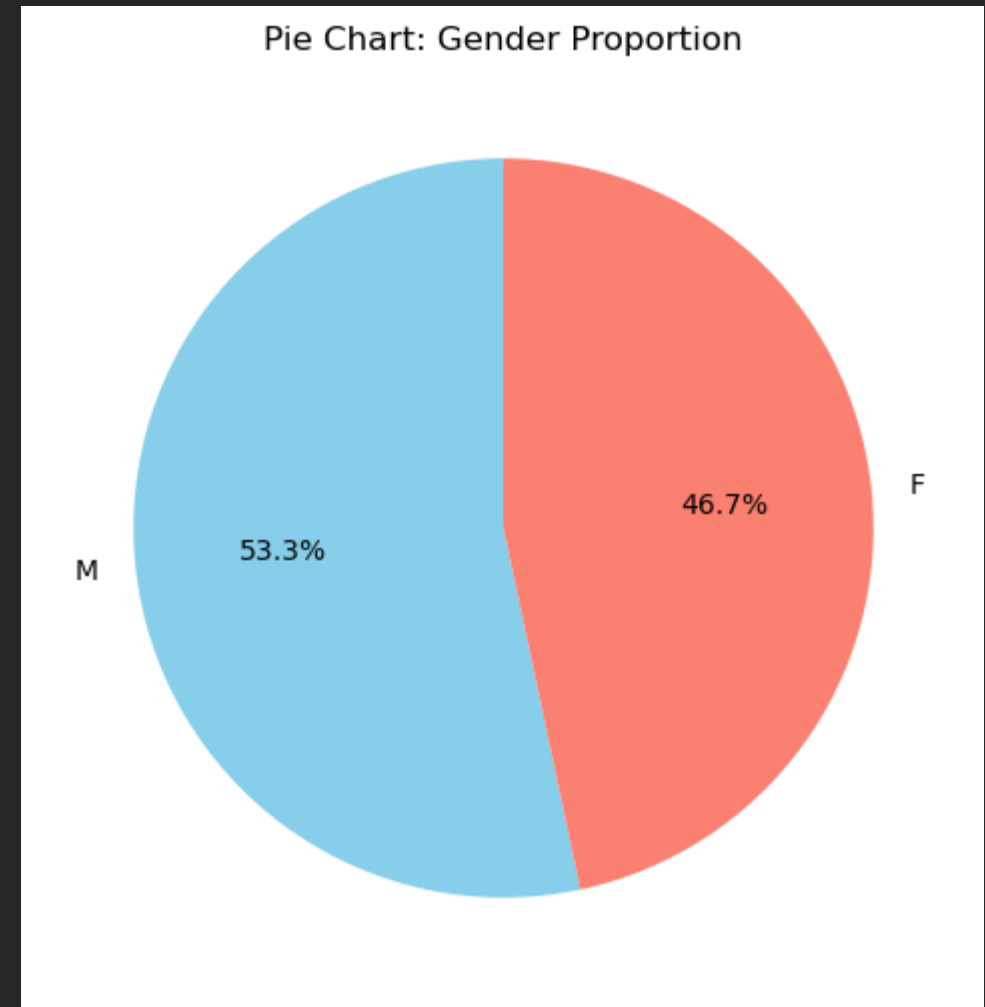
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Proportion and Percentage

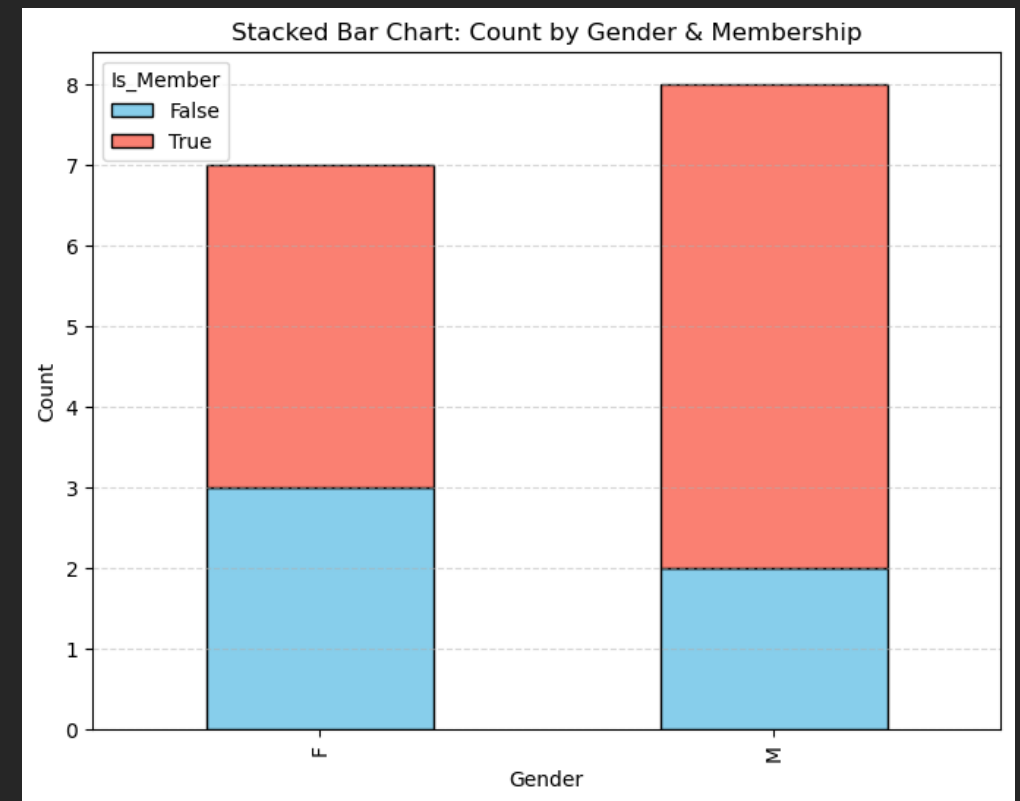
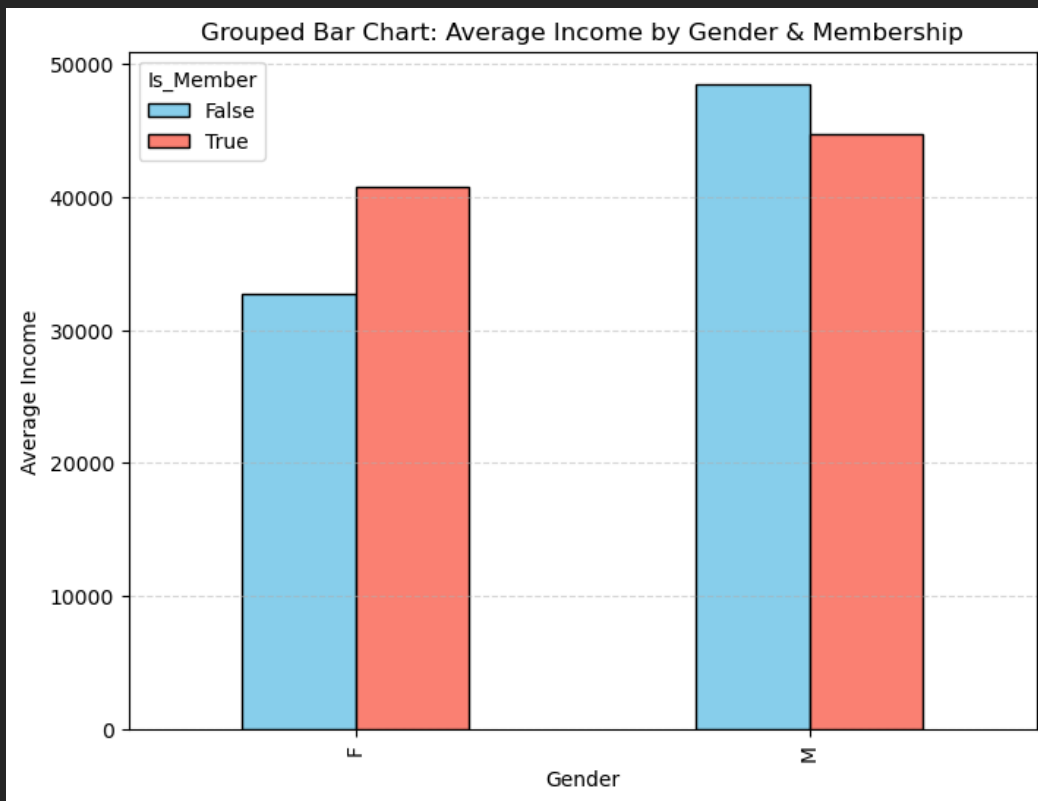
- Introduce Pie Chart

Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



1. Proportion and Percentage

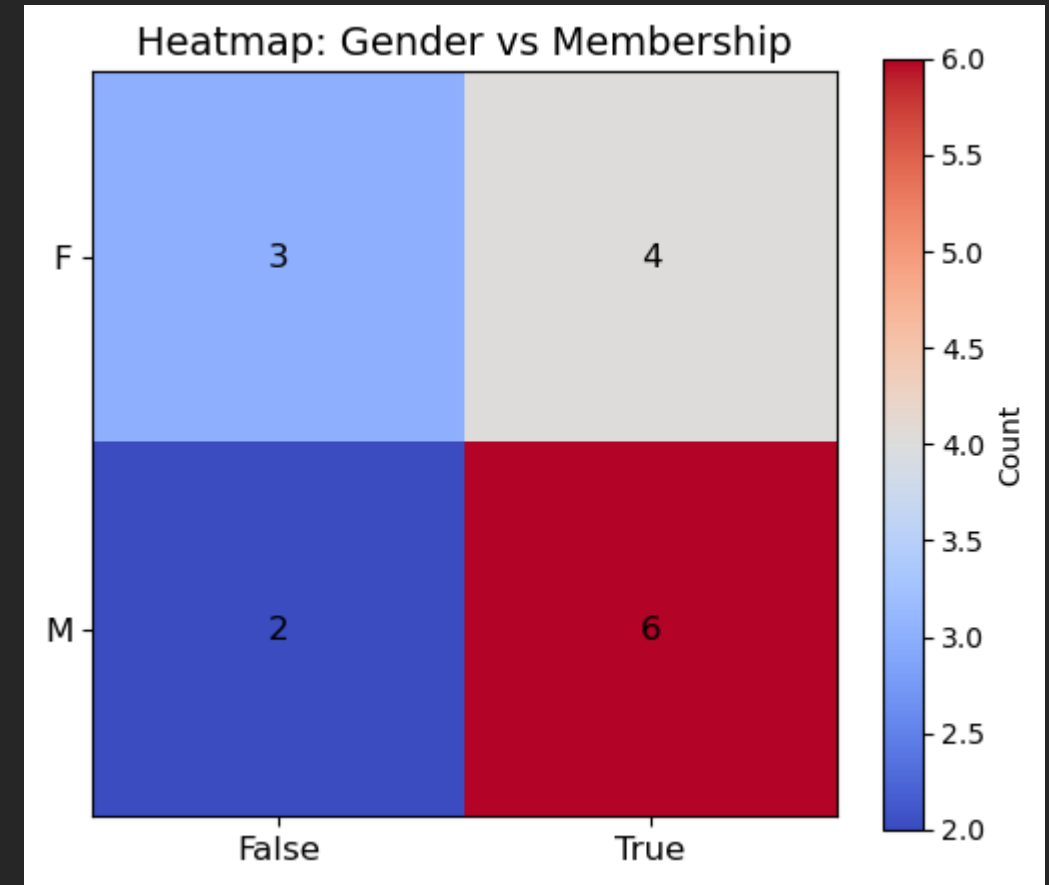
- Introduce grouped Bar Chart
- Introduce stacked Bar Chart



2. Cross-tabulation

- Cross-tabulation with Heatmap

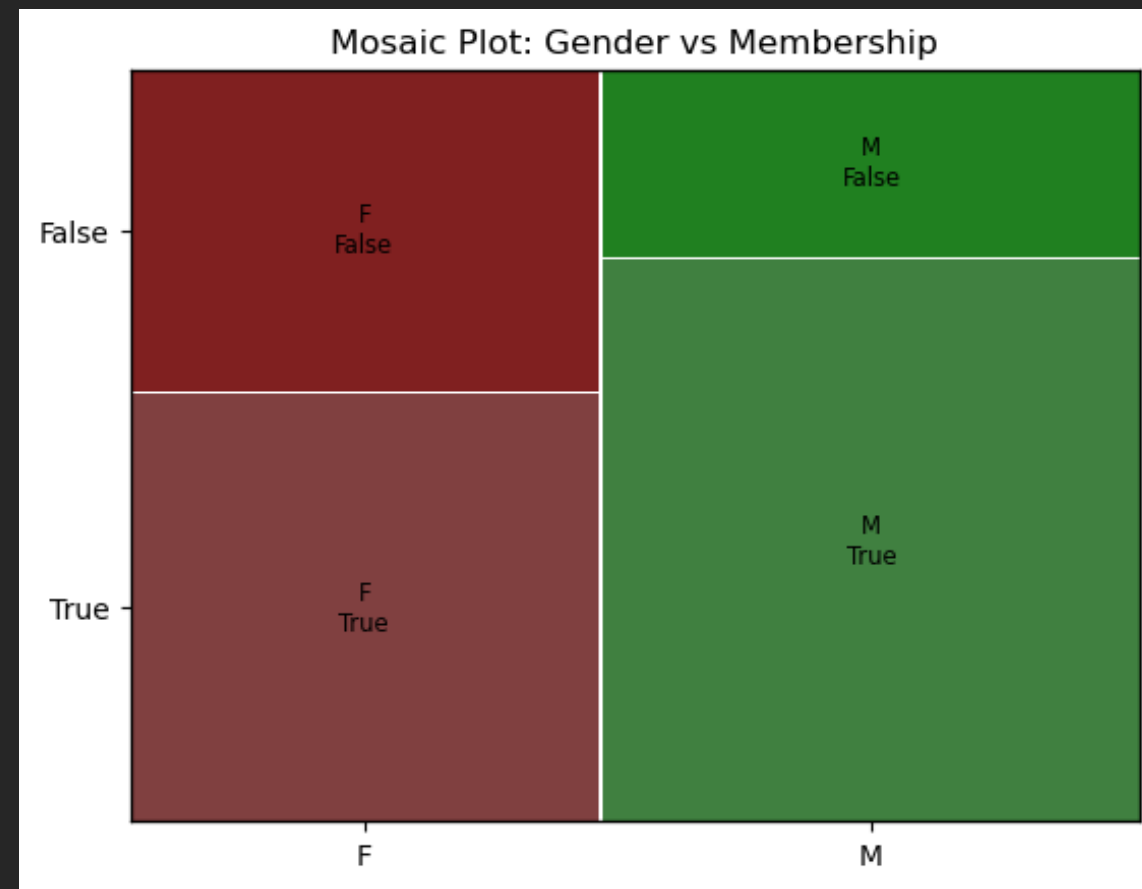
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE



2. Cross-tabulation

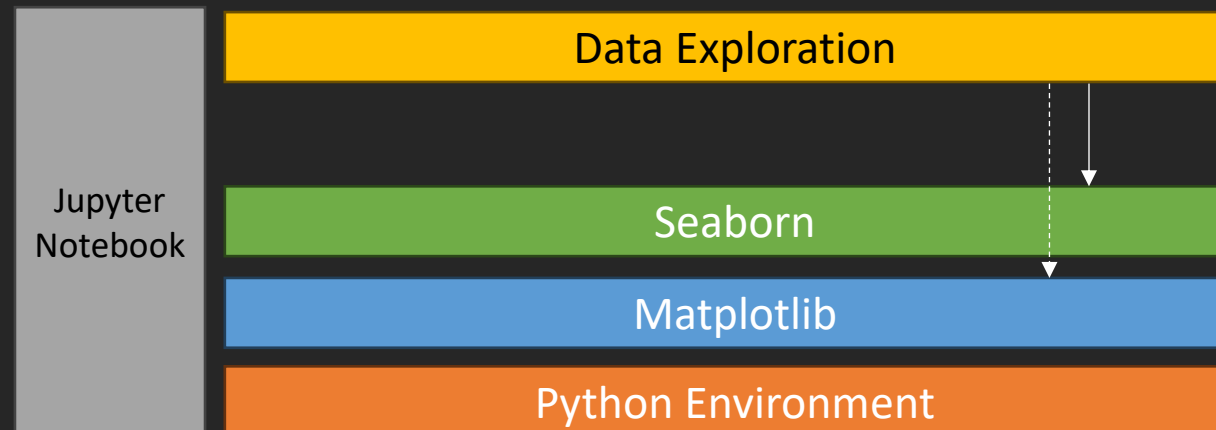
- Introduce Mosaic Chart

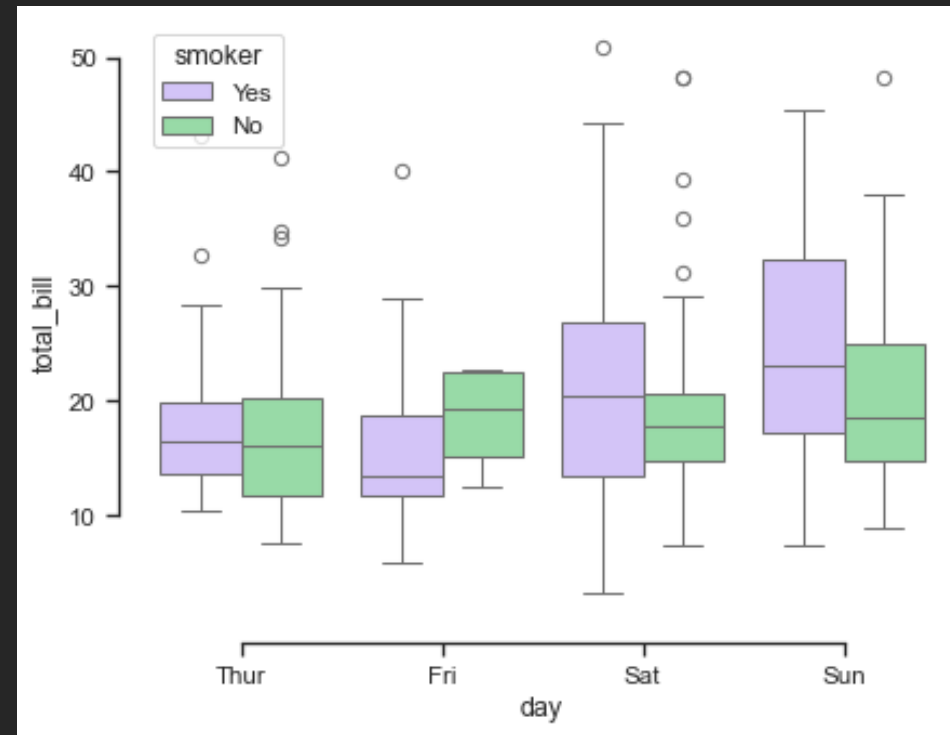
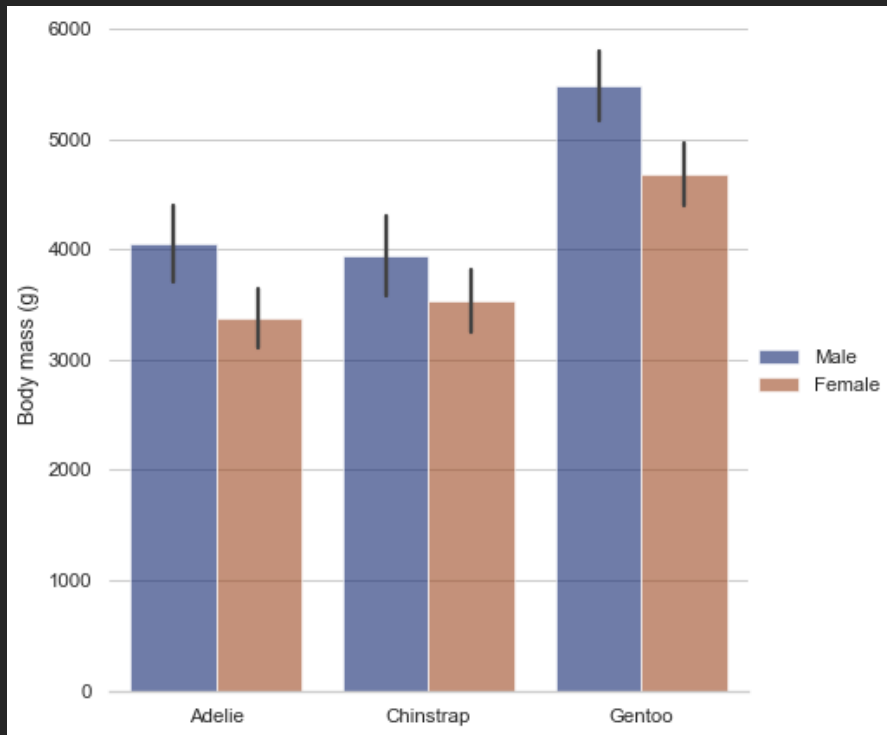
Cust_ID	Age	Income	Spend_SCR	Gender	Join_Date	Is_Member
1	25	25000	60	F	1/31/2020	TRUE
2	32	32000	70	M	2/29/2020	TRUE
3	40	45000	55	M	3/31/2020	FALSE
4	28	28000	65	M	4/30/2020	TRUE
5	35	35000	75	F	5/31/2020	FALSE
6	45	60000	80	M	6/30/2020	TRUE
7	50	72000	90	M	7/31/2020	TRUE
8	29	30000	50	F	8/31/2020	FALSE
9	31	31000	68	F	9/30/2020	TRUE
10	38	40000	72	M	10/31/2020	TRUE
11	42	52000	85	M	11/30/2020	FALSE
12	27	27000	58	F	12/31/2020	TRUE
13	33	33000	66	F	1/31/2021	FALSE
14	36	36000	74	M	2/28/2021	TRUE
15	48	80000	95	F	3/31/2021	TRUE

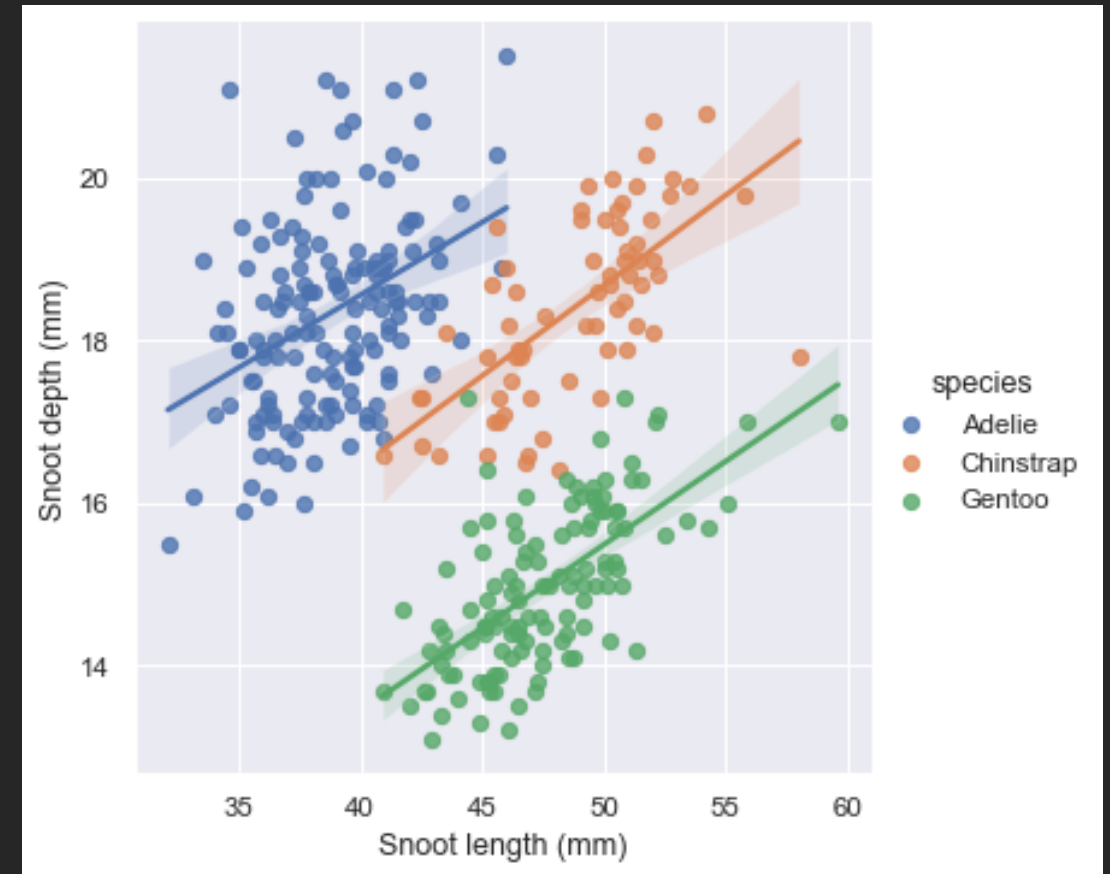
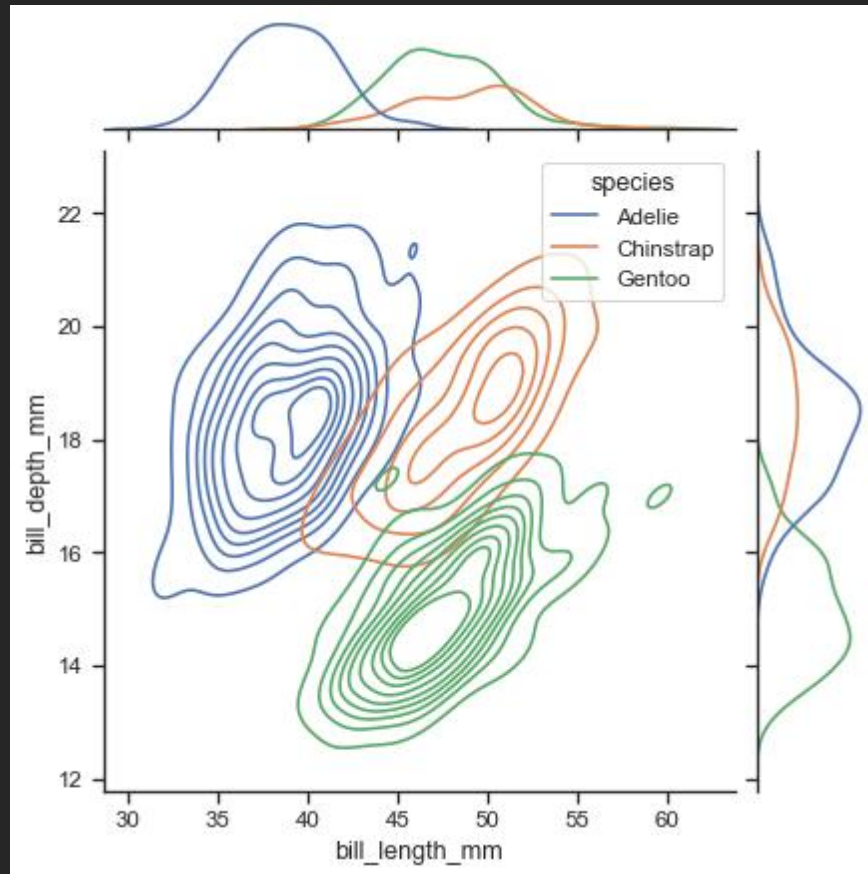


Using Seaborn Library

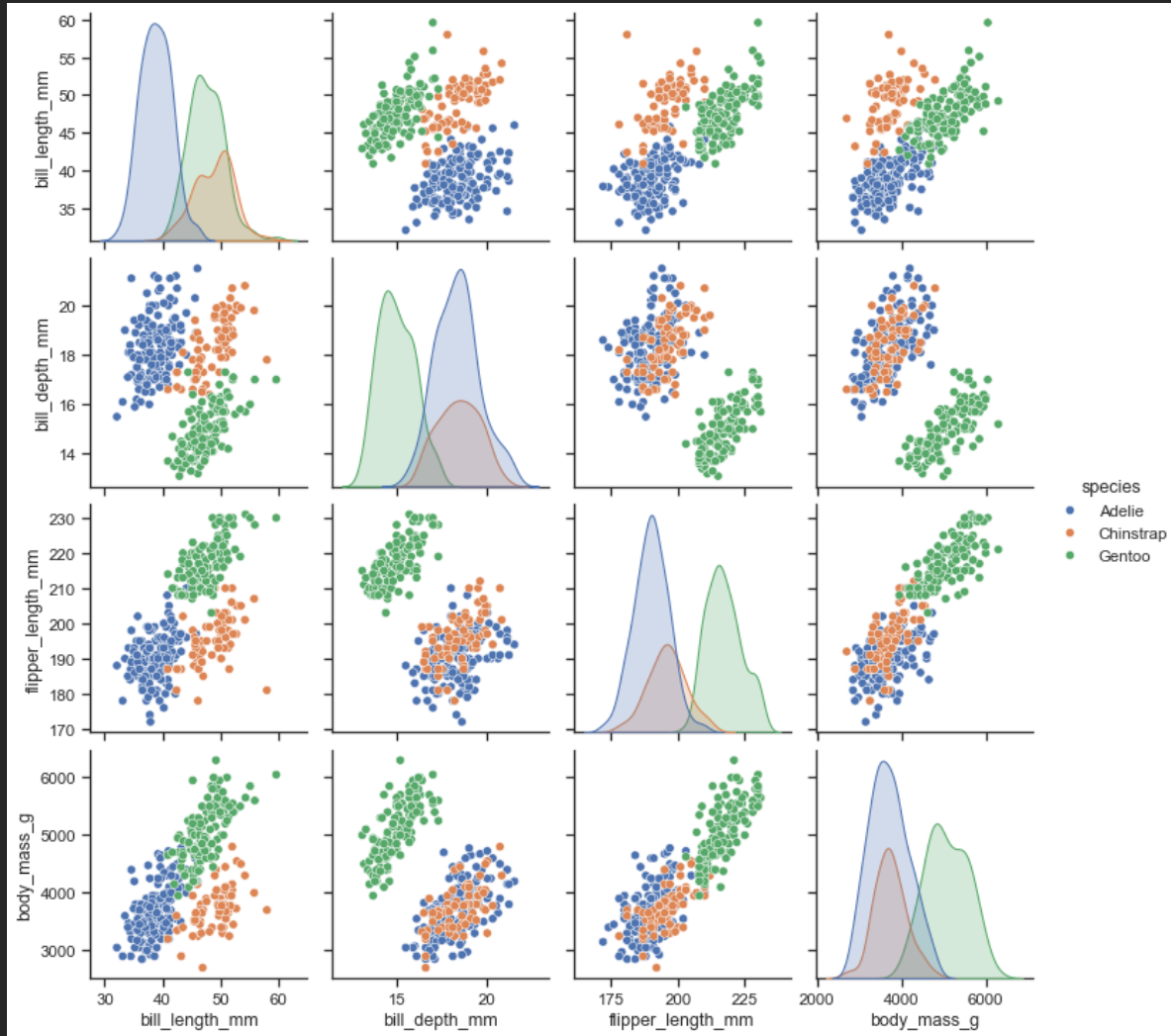
- Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.



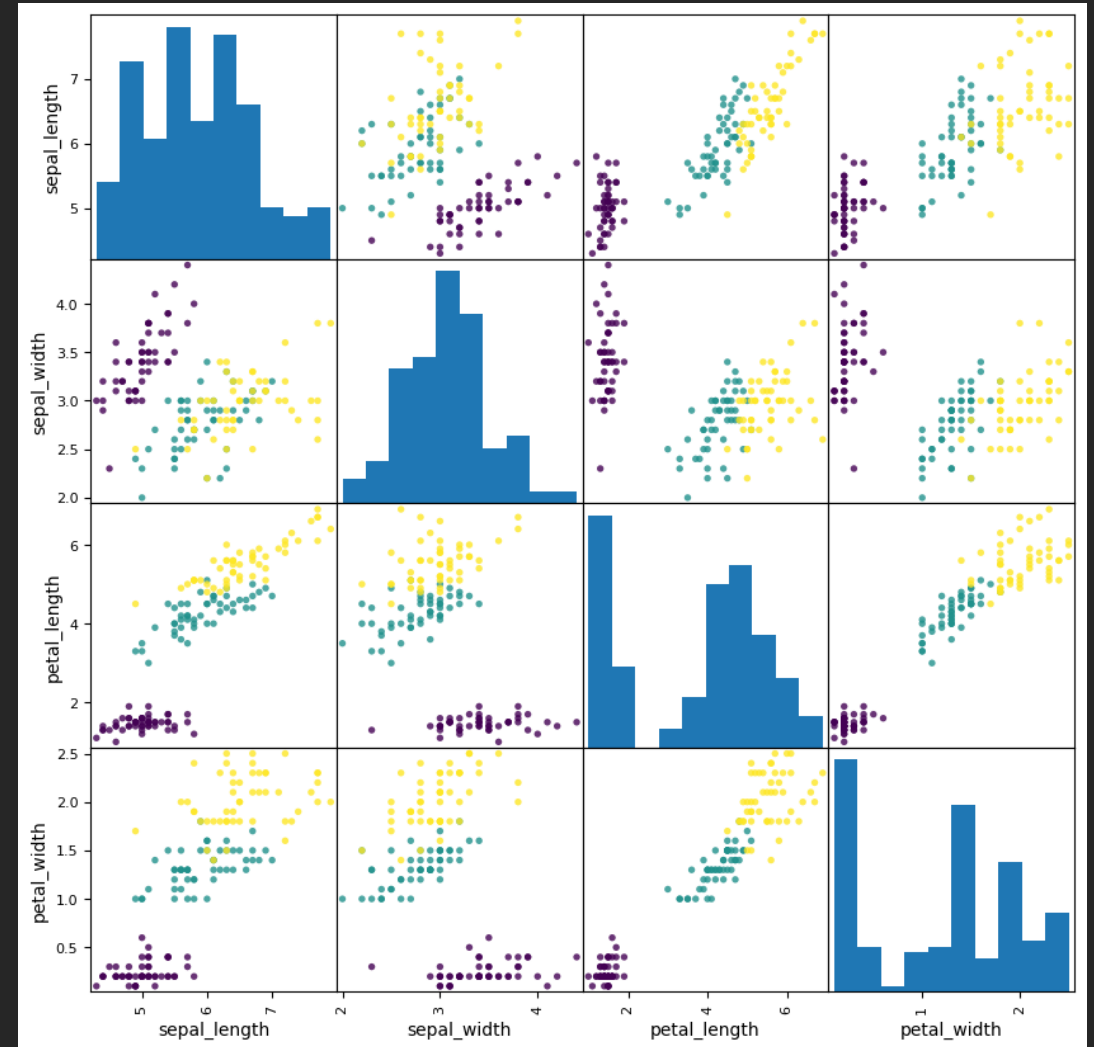




Seaborn



Matplotlib



Workshop : Analyze and Visualize Your Customer



Workshop : Analyze and Visualize Your Customer (Cont.)

Fields		Description
No	Identifier	
AcceptedCmp1		1 if customer accepted the offer in the 1st campaign, 0 otherwise
AcceptedCmp2		1 if customer accepted the offer in the 2nd campaign, 0 otherwise
AcceptedCmp3		1 if customer accepted the offer in the 3rd campaign, 0 otherwise
AcceptedCmp4		1 if customer accepted the offer in the 4th campaign, 0 otherwise
AcceptedCmp5		1 if customer accepted the offer in the 5th campaign, 0 otherwise
Age		Age
Complain		1 if customer complained in the last 2 years
Customer_Days		date of customer's enrollment with the company
Education		customer education level
Income		customer's yearly household income
Kidhome		number of small children in customer's household
Marital		customer's marital status

Workshop : Analyze and Visualize Your Customer (Cont.)

Fields	Description
MntFishProducts	amount spent on fish products in the last 2 years
MntFruits	amount spent on fruits in the last 2 years
MntGoldProds	amount spent on gold products in the last 2 years
MntMeatProducts	amount spent on meat products in the last 2 years
MntSweetProducts	amount spent on sweet products in the last 2 years
MntWines	amount spent on wines in the last 2 years
NumCatalogPurchases	number of purchases made using catalogue
NumDealsPurchases	number of purchases made with discount
NumStorePurchases	number of purchases made directly in stores
NumWebPurchases	number of purchases made through company's web site
NumWebVisitsMonth	number of visits to company's web site in the last month
Recency	number of days since the last purchase
Response	1 if customer accepted the offer in the last campaign, 0 otherwise
Teenhome	number of teenagers in customer's household

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
1	Load file with pandas Load the dataset from "ifood.xlsx" and check the number of rows and columns in the DataFrame.
2	Basic income statistics Calculate the mean, median, and mode of the "Income" column.
3	Age distribution Create a histogram of the "Age" column with gridlines and clear labels.
4	Range and IQR Compute the range and interquartile range (IQR) of the "MntWines" column.
5	Income grouping Use "pd.cut" to divide "Income" into 4 ranges and count the number of customers in each group.

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
6	Meat purchase analysis Compute the average spending on meat products "MntMeatProducts" grouped by education level "Education".
7	Groupby usage Use GroupBy to calculate the average "Income" by marital status "Marital".
8	Response proportion Calculate the proportion of customers who responded to the latest campaign "Response = 1"
9	Bar chart of marital status Create a bar chart showing the number of customers in each marital status category "Marital" using Matplotlib.
10	Correlation income vs wine Calculate the Pearson correlation between "Income" and "MntWines".

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
11	Boxplot income by education Create a boxplot of "Income" grouped by education level "Education".
12	Skewness and kurtosis Calculate the skewness and kurtosis of the "MntGoldProds" column.
13	Pairplot of purchases Create a pairplot of "MntWines", "MntMeatProducts", and "MntSweetProducts" to explore relationships.
14	Customers with children Filter customers with children "Kidhome" + "Teenhome" > 0 and compute their average "Income"
15	Top wine buyers Find the top 10 customers with the highest spending on wine "MntWines".

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
16	Income quantiles with qcut Use "pd.qcut" to divide "Income" into 4 quantiles and count customers in each group.
17	Stacked bar chart Create a stacked bar chart showing the number of customers by marital status "Marital" and education "Education".
18	Online purchase analysis Calculate the average "NumWebPurchases" and "NumWebVisitsMonth" grouped by marital status "Marital".
19	Multiple campaign responses Create a new column summing "AcceptedCmp1–5" and identify customers who accepted at least 2 campaigns.
20	Heatmap of correlations Create a heatmap of the correlation matrix for product spending variables "MntWines", "MntMeatProducts", "MntSweetProducts", "MntGoldProds".

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
21	Income vs Wine Spending Analyze whether higher `Income` is associated with higher spending on wine "MntWines". What insight can be drawn about premium product targeting?
22	Education vs Meat Purchases Compare average spending on meat "MntMeatProducts" across different "Education" levels. What does this suggest about dietary preferences by education?
23	Marital Status vs Online Behavior Examine differences in "NumWebPurchases" and "NumWebVisitsMonth" across "Marital" categories. What insight can be drawn about digital engagement by family status?
24	Age vs Campaign Response Investigate whether age "Age" influences the likelihood of responding to campaigns "Response". Which age group is most receptive?
25	Children in Household vs Spending Compare average spending on sweets "MntSweetProducts" between households with children "Kidhome" + "Teenhome" > 0 and those without. What does this reveal about family consumption patterns?

Workshop : Analyze and Visualize Your Customer (Cont.)

No	Exercises
26	High-value Customers Identify customers with the highest combined spending "MntWines" + "MntMeatProducts" + "MntGoldProds". What characteristics (age, marital, education) do they share?
27	Campaign Acceptance Patterns Analyze customers who accepted multiple campaigns "AcceptedCmp1–5". What demographic or behavioral traits distinguish them?
28	Recency vs Spending Explore the relationship between "Recency" (days since last purchase) and total spending. Do recent buyers tend to spend more?
29	Income Segmentation Segment customers into income quartiles (pd.qcut) and compare their product preferences (wine vs meat vs gold). What insights emerge about product positioning?
30	Correlation Heatmap Create a correlation heatmap of product spending variables "MntWines", "MntMeatProducts", "MntSweetProducts", "MntGoldProds"). Which products are often purchased together, and what cross-selling opportunities exist?

Chapter's conclusions:

- ❑ Students learn to summarize and interpret data distributions using descriptive statistics and simple plots.
- ❑ Students understand how to measure and visualize relationships between variables through correlation and graphical techniques.
- ❑ Students gain the ability to analyze categorical data and compare groups effectively using comparative visualizations.